Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

Residence Information
US Residency
Fort Collins, CO
UNITED STATES

Mailing Address
149 Remington Street Unit 425
Fort Collins, CO 80524
UNITED STATES

Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**AUTONOMOUS API CONNECTOR SYNTHESIS WITH SPECIFICATION DISCOVERY, ONTOLOGY MAPPING, AND DATA LOSS PREVENTION ENFORCEMENT**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

HeadySystems Inc.
149 Remington Street Unit 425
Fort Collins, CO 80524
UNITED STATES

9702000327

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eric@headyconnection.org

Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	03/03/2026

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

AUTONOMOUS API CONNECTOR SYNTHESIS WITH SPECIFICATION DISCOVERY, ONTOLOGY MAPPING, AND DATA LOSS PREVENTION ENFORCEMENT

Title of the Invention	Autonomous API Connector Synthesis with Specification Discovery, Ontology Mapping, and Data Loss Prevention Enforcement
Docket Number	HS-2026-007
Inventor(s)	Eric Haywood
Residence(s)	149 Remington Street Unit 425, Fort Collins, CO 80524
Correspondence Address	HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524
Attorney/Agent (Name & Reg. No.)	None
U.S. Government Interest (Yes/No)	No
Gov. Agency & Contract No.	N/A

Title of the Invention

[0001] Autonomous API Connector Synthesis with Specification Discovery, Ontology Mapping, and Data Loss Prevention Enforcement

Cross-Reference to Related Applications

[0002] This application claims the benefit of priority and is related to co-pending provisional applications filed concurrently herewith by the same inventor and assignee.

Field of the Invention

[0003] This invention relates to systems and methods for automatically discovering external API specifications, generating type-safe connector code, enforcing data loss prevention policies, and supporting runtime protocol switching between REST, WebSocket, SSE, gRPC, and GraphQL.

Background of the Invention

[0004] Current API integration requires manual development: reading API documentation, writing client code, handling authentication, managing protocol differences and ensuring compliance with data protection policies. This process is labour-intensive, error-prone and fails to adapt when API specifications change. No existing system autonomously discovers an API's specification, maps its data model to an internal ontology, generates safe connector code, and enforces DLP policies—without human intervention.

Summary of the Invention

[0005] The invention provides a system and method as described herein.

[0006] Discovery: Automatically fetches and parses OpenAPI, Swagger and GraphQL specifications from a target URL, extracting endpoints, authentication requirements, data schemas and protocol capabilities.

[0007] Ontology Mapping: Maps discovered API data structures to an internal semantic ontology, establishing bidirectional correspondences between external schemas and the system's canonical data model.

[0008] Code Generation: Synthesizes type-safe connector code with embedded authentication handling, error recovery and rate limiting, reducing manual development effort.

[0009] DLP Enforcement: Applies configurable data loss prevention rules to all generated connector code, blocking egress of personally identifiable information (PII), credentials and protected intellectual property.

[0010] Protocol Switching: Enables runtime switching between REST, WebSocket, SSE, gRPC and GraphQL protocols without regenerating connector code.

Detailed Description of the Preferred Embodiment

A. Discovery Engine

[0011] The discovery engine uses pattern-based URL resolution to fetch API specifications. It appends candidate paths (such as /openapi.json, /swagger.json and /.well-known/openapi) and parses the specification to extract endpoints with HTTP methods, parameter schemas, authentication mechanisms, rate-limiting policies and supported communication protocols.

[0012] Machine-readable API specifications (like OpenAPI and Swagger) allow automatic generation of client SDKs and server stubs. They provide schema validation and improve documentation quality, enabling downstream automation of client code and test generation.

B. Ontology Mapper

[0013] The ontology mapper establishes field-level mappings between external API data structures and the system's canonical data model. It supports direct 1:1 mappings, computed mappings derived from multiple source fields and enumeration mappings for translating values across domains.

C. Code Generator and Security Linter

[0014] The code generator produces JavaScript connector code with embedded authentication handling, error recovery and rate limiting. A security linter scans the generated code against unsafe patterns, such as eval, Function constructors and innerHTML assignments, rejecting connectors that contain unsafe code. Generated code is quarantined if it violates security policies.

[0015] After generation, data loss prevention rules are applied to all outbound data flows. Configurable rules block egress of email addresses and phone numbers, authentication credentials like API keys or tokens and designated intellectual property files (e.g., .als, .wav, .aif, .flac, .mid).

D. Protocol Switching

[0016] The connector includes a transport-agnostic interface that can switch between REST, WebSocket, Server-Sent Events, gRPC and GraphQL at runtime. This allows dynamic adaptation to service capabilities and network conditions without regenerating connector code.

Claims

1. A computer-implemented method for autonomous API connector synthesis, comprising: (a) discovering an API specification by fetching and parsing a structured specification document from a target URL; (b) mapping the discovered API's data structures to an internal ontology by establishing field-level correspondences; (c) generating type-safe connector code with embedded authentication, error recovery and rate limiting; (d) enforcing data loss prevention rules on all data flowing through the generated connector; and (e) enabling runtime protocol switching between at least two of REST, WebSocket, SSE, gRPC and GraphQL without regenerating connector code.

2. The method of claim 1, wherein the data loss prevention enforcement comprises blocking egress of personally identifiable information, authentication credentials and designated intellectual property file types.

3. The method of claim 1, wherein the generated connector code is scanned against a set of unsafe code patterns and quarantined if any unsafe pattern is detected.

4. A system for autonomous API integration comprising: a specification discovery engine that parses OpenAPI, Swagger and GraphQL specifications; an ontology mapper that establishes bidirectional field mappings; a code generator that produces connector code with embedded security analysis and DLP enforcement; and a protocol switching engine that enables runtime transport protocol changes.

Implementation Reference

[0017] The Dynamic Connector Service may be implemented in the file `src/services/dynamic-connector-service.js` using the class `DynamicConnectorService`. Connectors enforce three categories of DLP rules: no PII egress, no credentials, and no music intellectual property. Protocol switching between REST, WebSocket, SSE, gRPC and GraphQL is supported.

References

[1] OpenAPI and Swagger specifications enable automatic generation of client SDKs, server stubs and tests, improving validation and documentation【679056392993234†L411-L417】

[2] OpenAPI specifications allow code generation of client SDKs and server stubs, reducing manual effort【679056392993234†L514-L518】

[3] Data loss prevention solutions prevent sensitive information from being lost or exfiltrated, protecting PII and intellectual property【645251413980947†L152-L169】

UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/995,320	**03/03/2026 02:12:26 PM Z ET**	**-**

Title of Invention

AUTONOMOUS API CONNECTOR SYNTHESIS WITH SPECIFICATION DISCOVERY, ONTOLOGY MAPPING, AND DATA LOSS PREVENTION ENFORCEMENT

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	4822	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74697606	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74697606.pdf	5	Application Data Sheet	109 KB
HS-2026-007-Final-Assigned-Updated-APP.TEXT.docx	4	Application body structured text document	39 KB

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
generatedADS74697606.pdf	5686C5822762CF82E7CC57ECCAF0AE8CEC7B3D1225DB1898

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8C82604F7B43349B918BFF1E9FE186FB

HS-2026-007-Final-Assigned-Updated-APP.TEXT.docx

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39DE684AD56751D23B1BEDC348A579

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.